Outline for Prevention and Lifecycle Health Types

November 10, 2026

1 Introduction

Question: what are the returns to prevention over the life course, for whom, and when?

Key results:

- Return to prevention depends on each lifecycle health type's mean path, not on the cross-section
- Divergence then convergence over the lifecycle
- Unhealth group of ≈20% that carries most of the targetable return
- Health is by far the most important observable for targeting
- Target the unhealthy when they are young and the healthy when then are old
- Uniform prevention could increase lifespan an health gradients

Related literature:

- Literature on prevention
- Earnings dynamics and its use of the covariance matrix.
- Health measurement in economics (factor and item-response models).
- Stochastic specification closest to Hosseini et al. (2022): log frailty as fixed effect plus AR(1) plus transitory shock.

2 Conceptual framework

2.1 Setup: deterministic type and stochastic noise

- Health of person i of type j at age a:

$$h_{i,a} = \bar{h}_{j,a} + u_{i,a}, \qquad \bar{h}_{j,a} \equiv H_j + \sum_{a'=1}^{a} d_{j,a'}. \tag{1}$$

- Three kinds of information: what *stays* (H), what is *in the moment* (u), how one is *evolving* (d).
- Prevention targets the third; the first two are paid for as treatment.
- Each leaves a different trace on the covariance matrix: the basis for identification.
- Types are the object of interest because the return to prevention depends on a type's mean path.
- Linear illustration ($d_{j,a} = d_j$):

$$\begin{aligned} \bar{h}_a &= \bar{H} + a\bar{d}, \\ \mathrm{Var}(h_a) &= V_H + 2aC_{Hd} + a^2 V_d + v_a, \\ \mathrm{Cov}(h_s, h_t) &= V_H + (s+t)C_{Hd} + st\,V_d + \mathrm{Cov}(u_s, u_t). \end{aligned} \tag{2}$$

- Units matter: the covariance matrix treats health as cardinal, so we need an anchor (costs).

2.2 Policy: the treatment bill, and prevention as avoiding it

- Prevention from r at intensity μ scales every future mean change by $(1-\mu)$:

$$\bar{h}^{\mu}_{j,a} = \bar{h}_{j,a} + \mu\,(\bar{h}_{j,r-1} - \bar{h}_{j,a}) \quad \text{for } a \geq r.$$

- We assume prevention acts on health dispersion that is (i) deterministic and (ii) actionable.
- Treatment bill still to come from age r:

$$C(r) = \sum_{a\geq r} \mathbb{E}[c(h_{i,a})] = \sum_{a\geq r} \Big[\underbrace{c(\bar{h}_a)}_{\text{mean path}} + \underbrace{\textstyle\sum_j \pi_{j,a} c(\bar{h}_{j,a}) - c(\bar{h}_a)}_{\text{type differences}} + \underbrace{\mathbb{E}[c(h_{i,a})] - \textstyle\sum_j \pi_{j,a} c(\bar{h}_{j,a})}_{\text{realised deviations}} \Big]. \quad (3)$$

  The third part is transitory and never prevented.
- Illustrative cost $c(h) = (1-h)^2$. Population cost at age a:

$$\mathbb{E}[c(h_{i,a})] = (1-\bar{h}_a)^2 + \mathrm{Var}_j(\bar{h}_{j,a}) + \textstyle\sum_j \pi_j \mathrm{Var}(u_{i,a} \mid j). \quad (4)$$

- In which case the total bill from age r onwards is

$$C(r) = \sum_{a\geq r} \big[(1-\bar{h}_a)^2 + \mathrm{Var}(h_{i,a})\big] \quad (5)$$

- Returns to prevention are priced as the treatment costs that the prevention avoids.

Result 1 (Return to prevention by type)**.**

$$P_j(r) = 2\sum_{a\geq r} \underbrace{(1-\bar{h}_{j,a})}_{\textit{gap at } a} \underbrace{(\bar{h}_{j,r-1} - \bar{h}_{j,a})}_{\textit{preventable decline by } a} \equiv \sum_{a\geq r} s_j(a \mid r). \quad (6)$$

A function of the type's mean path alone. A type pays back most where it is already in poor health and still declining.

Corollary 1 (The uniform policy)**.**

$$P(r) = 2\sum_{a\geq r} \Big[\underbrace{(1-\bar{h}_a)(\bar{h}_{r-1} - \bar{h}_a)}_{\textit{mean channel}} + \underbrace{\mathrm{Cov}_j\big(1-\bar{h}_{j,a},\, \bar{h}_{j,r-1} - \bar{h}_{j,a}\big)}_{\textit{between-type channel}} \Big]. \quad (7)$$

The between-type channel splits into pathway, persistence and accumulation. The mean channel is most of the uniform return; types add the dispersion of $P_j(r)$, which brings in targeting.

- General convex cost: marginal cost times decline,

$$P_j(r) = \sum_{a\geq r} G_{j,a}\,(\bar{h}_{j,r-1} - \bar{h}_{j,a}), \qquad G_{j,a} \equiv \mathbb{E}[-c'(h_{i,a}) \mid j]. \quad (8)$$

- Prevention acts on change, so the policy needs $(\pi_j, \bar{h}_{j,\cdot})$; the covariance structure is needed only to separate path from noise and to recover a person's type.
- $P(r)$ is first order and can exceed the bill. For the cubic cost we estimate, the saving at any μ is give by a third order approximation, exact.

Result 2 (Targeting)**.**

- *Any allocation $\mu(z)$ with the same mean saves* $\mathrm{Cov}(\mu(z), \mathbb{E}[P_i \mid z])$ *over uniform: a signal is worth the return dispersion it reveals.*
- *Targeting on initial health works through the first row of the covariance matrix, attenuated by the signal share; targeting on type gains* $\max_j P_j(r) - \sum_j \pi_j P_j(r)$.

2.3 Identification

- Observed covariance = between-type part + noise part; the returns need the first, so we need to separate it from the second.
- Assumptions: A1 window linearity, A2 noise orthogonal to (H, d), A3 a noise class. Linear case:

$$\mathrm{Cov}(h_s, h_t) = V_H + (s+t)C_{Hd} + st\,V_d + \mathrm{Cov}(u_s, u_t). \tag{9}$$

- Unrestricted noise variances can never rule out pure noise; stationary noise identifies (C_{Hd}, V_d) from the diagonal but not V_H from σ_u^2.
- The noise class below, a persistent AR(1) component plus a one-period shock, is the persistent–transitory specification (standard in earnings and income dynamics and also used in health).
- The transitory shock includes both pure noise (measurement error) and genuine changes in health that are gone by the next period; the covariance matrix cannot tell them apart.

Proposition 1 (Identification within the persistent–transitory class)**.** *Under A1–A2 and* $\mathrm{Cov}(u_s, u_t) = \rho^{t-s} v_s + \mathbf{1}\{s=t\}\sigma^2_{\varepsilon,s}$*: five waves identify* V_H, C_{Hd}, V_d, ρ *and both noise variances; four suffice without the transitory shock or with stationary noise; three without it and with stationary variance; two if also* $C_{Hd} = 0$ *and memoryless noise.*

- One identifying force per region of the matrix: **diagonal** (C_{Hd}, V_d from differences); **first row** (a line cannot mimic a decay: splits level from noise, gives ρ; the transitory shock is the anchor's excess); **interior** (V_d, overidentification).
- Linearity is local, so $C_{Hd}(a)$ can change sign: a flattening variance profile requires it. V_H at a given age is an accumulation of past dispersion.
- The uniform return is priced by the surface's moments alone:

$$P(r) = 2\sum_{a \geq r} \big[(1-\bar{h}_a)(\bar{h}_{r-1} - \bar{h}_a) + (a-r+1)\,C_{Hd} + a(a-r+1)\,V_d\big]. \tag{10}$$

- C_{Hd}, V_H and V_d are all informative for the value of timing and targeting.

Policy object	Moments it needs
$C(r)$, and the mean channel of $P(r)$	mean path and variance: Figure 2(a), (b)
accumulation margin; timing $r \mapsto P(r)$	V_d (with C_{Hd}): curvature of the surface
pathway margin; targeting on a level	$C_{Hd}(a)$, its sign: Figure 2(c)
targeting on a type	$P_j(r)$ across types: Section 4

3 Data and motivating descriptives

3.1 Data and the health measure

- UKHLS waves 1–15 (2009–24), ages 20–90. Lifecycle sample: at least three observed ages, 334,194 person-ages on 38,963 people.
- Eight physical-health items, none tied to employment.
- Graded-response model (Samejima, 1969):

$$P(Y_j \geq c \mid \theta) \;=\; \frac{1}{1+\exp\{-a_j(\theta - b_{j,c-1})\}} \tag{11}$$

- Latent health θ is assumed standard normal in the pooled sample
- Bounded expected-score twin:

$$h(\theta) \;=\; \frac{\sum_{j=1}^{J}\sum_{c=2}^{k_j} P(Y_j \geq c \mid \theta)}{\sum_{j=1}^{J}(k_j - 1)}, \qquad h_i = h(\hat{\theta}_i), \quad \hat{\theta}_i = E[\theta \mid \mathbf{y}_i]. \tag{12}$$

- Plus a 31-deficit frailty index and $\log(\text{frailty} + 1/31)$; h behaves like frailty, θ like log frailty.
- No measure is uniquely cardinal, but covariances are; costs give the anchor. Main text uses θ the others are in an Appendix.
- Advantage of θ - distinguishes between the healthy, not just the unhealthy which is important for lifecycle dynamics.

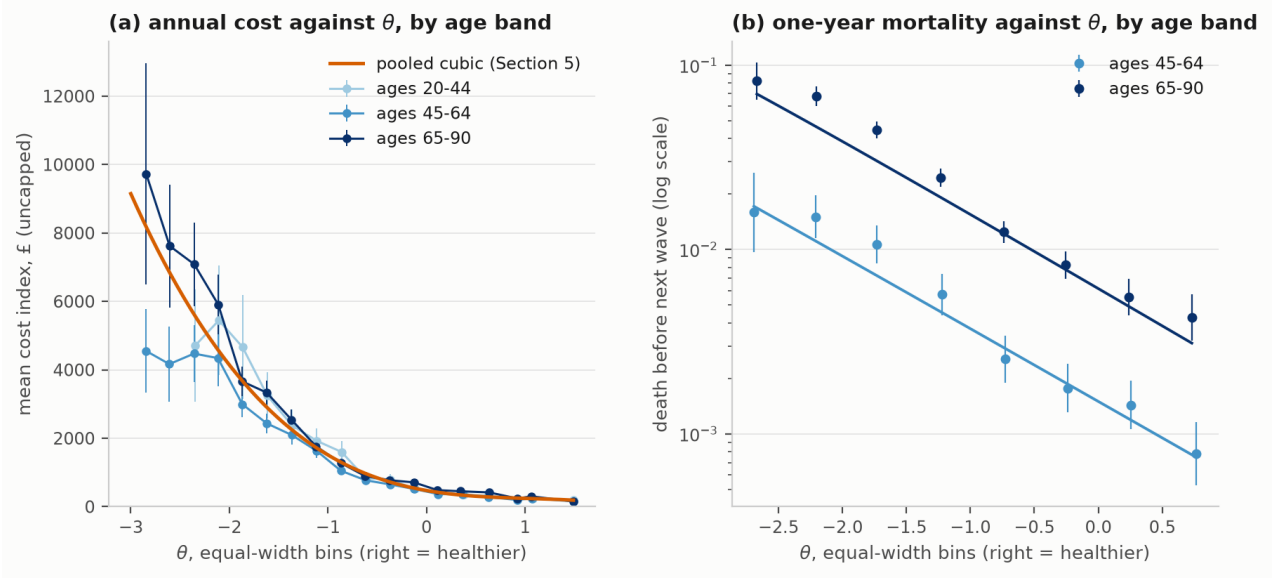


Figure 1: Annual cost (a) and one-year mortality (b) against θ, by age band: binned means with 95% intervals; lines in (b) are within-band logits.

- Cost proxy: annual NHS cost from in-patient, out-patient and GP use, waves 7–15, 2024/25 prices:

$$\widehat{C}_{it} \;=\; \underbrace{\text{adm}_{it}\left[\,c_{\text{short}}\,\mathbf{1}\{n_{it} \leq 1\} + c_{\text{bed}}\,n_{it}\,\mathbf{1}\{n_{it} \geq 2\}\right]}_{\text{in-patient}} \;+\; \underbrace{c_{\text{op}} \cdot \widehat{n}^{\text{op}}_{it}}_{\text{out-patient}} \;+\; \underbrace{c_{\text{gp}} \cdot \widehat{n}^{\text{gp}}_{it}}_{\text{GP}}. \tag{13}$$

- Wrong in level (£900 vs £3,000 a head), right in shape.
- Figure 1: cost is convex in poor health ($t = 17.4$ on the quadratic) and close to age-invariant above $\theta \approx -2$; mortality is roughly log-linear in θ, but age carries extra mortality risk.

3.2 The moments of health over the life course

First look at health covariance matrix:

- (a) The mean decline accelerates with age: on its own this says intervene late.
- (b) Variance rises 60% from 30 to 60, peaks at 62, then falls: $C_{Hd}(a)$ turns negative after 60.
- (c) A one-year transitory drop in every row, then divergence among the young and convergence among the old.

- One stationary noise process cannot produce both, so either noise moves with age or C_{Hd} changes sign. Implication: target the low early; later, targeting on levels pays little.
- Same on log frailty; bounded measures (h, frailty) keep variance growing after 60.

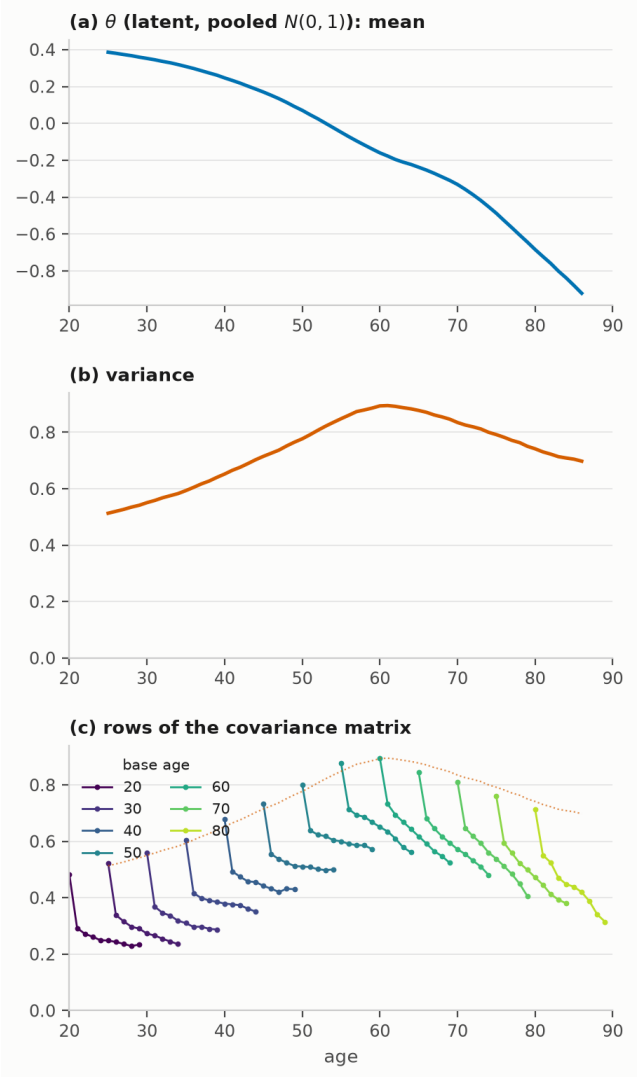


Figure 2: Model-free moments of θ by age: (a) mean, (b) variance, (c) rows of the covariance matrix leaving the diagonal at five-year base ages (dotted: the variance profile).

3.3 The linear approximation on real data

- Fit V_H, C_{Hd}, V_d and noise by constrained least squares on the ten moments of four ages four years apart, within four age bands and persistent–transitory stochastic process.
- $\mathrm{Corr}(H, d)$ goes from positive to negative with age (+0.45, +0.78, −0.56, −0.53); the sign path is robust across cases and measures.

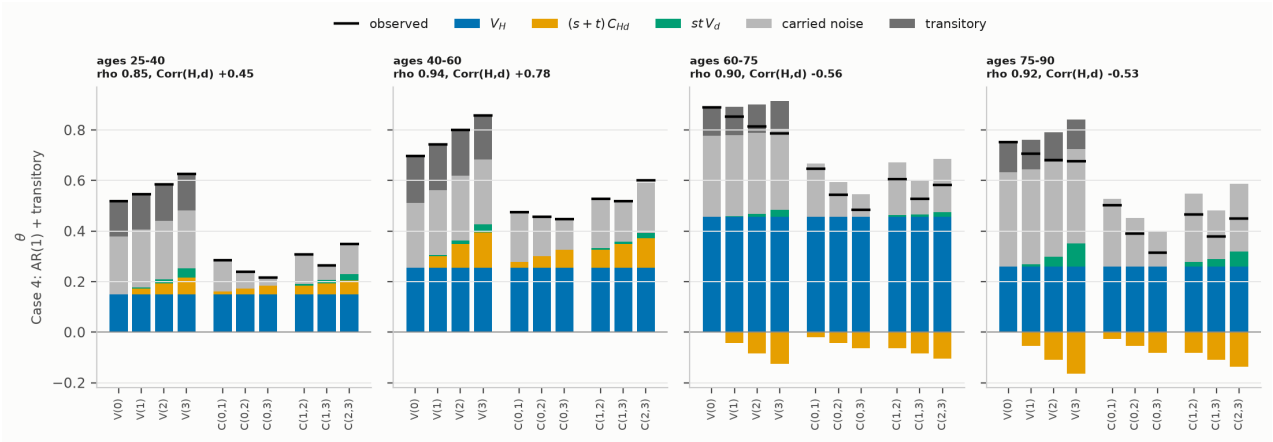


Figure 3: The linear approximation on θ under persistent–transitory stochastic process, by age band: bars stack the fitted layers of the variances $V(k)$ and covariances $C(k,l)$ of four ages four years apart; black ticks are the observed moments.

4 Lifecycle types

4.1 Why and how

- Health at 50 is an outcome of earlier dynamics, and divergence happens before late middle age, so we need the full lifecycle (most work uses the old or short windows).
- Cohort-independent types with cohort-dependent shares, estimated three ways: partial K-means, Bayesian mixtures, then against observables.
- Challenges: power and type definition; mortality and attrition; cohort effects; units and bounds.

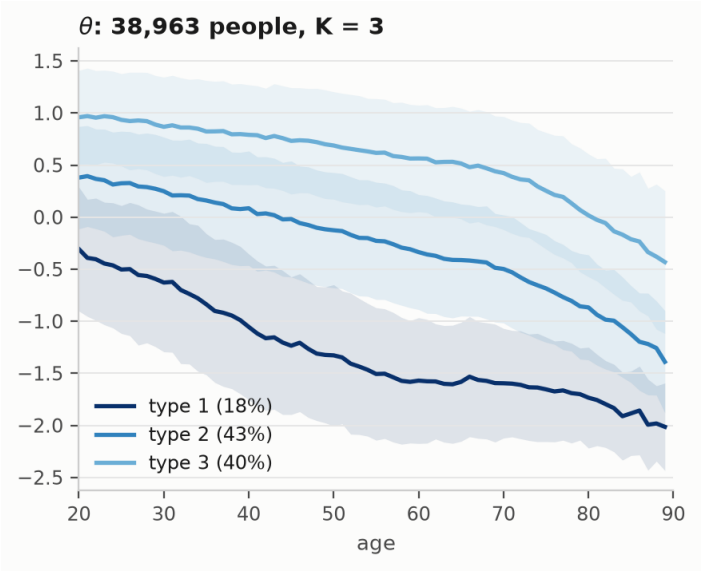


Figure 4: Partial K-means types ($K = 3$) on θ: the mean of θ by age in each type, shaded to ± 1 within-type sd. The legend gives each type's share of people.

4.2 Non-parametric types: partial K-means

- Partial K-means: minimise squared deviations over observed entries only with every person-wave weighted equally.
- Figure 4 shows $K = 3$ ordered paths, 18/43/40%. The worst type falls fastest to 60, then the order reverses: divergence then convergence.
- Within-type deviations are persistent (lag-1 autocorrelation 0.37–0.64), so the final model needs persistence.
- Share of population in worst type increases with age (opposite of mortality selection) so parameteric model should deal with this too.

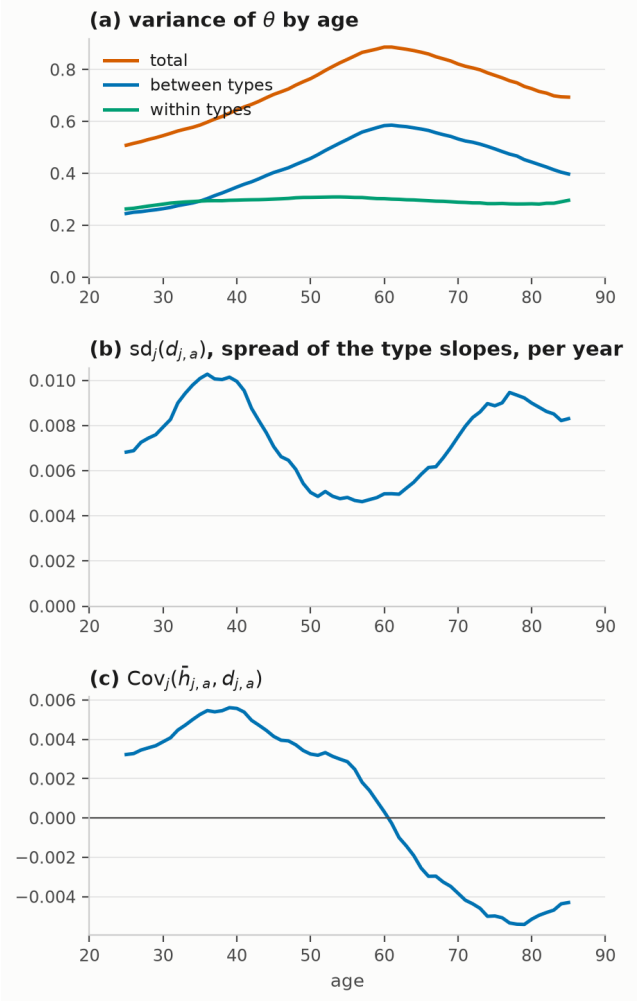


Figure 5: Between versus within the partial K-means types on θ, by age: (a) the variance split, (b) the spread of the type slopes, (c) the covariance of type levels and slopes.

- Figure 5 is analogue of Figure 2 with the cluster types
- Within type variance is relatively constant and variance in $d_{j,a}$ has U shape
- $\text{Cov}_j(\bar{h}_{j,a}, d_{j,a})$ changes sign at 61.

4.3 Parametric types: Bayesian mixtures

- Baseline: mixture of quadratic growth curves, $\tilde{a} = (a - 55)/10$, estimated in Stan:

$$y_{ia} = \alpha_{k_i} + \beta_{k_i}\tilde{a} + \gamma_{k_i}\tilde{a}^2 + \varepsilon_{ia}, \qquad \varepsilon_{ia} \sim N(0, \sigma^2) \text{ i.i.d.} \tag{14}$$

  The transitory-only specification; same typology as K-means (ARI 0.85, 95% same class).
- Main specification, persistent–transitory: an AR(1) state plus a transitory shock:

$$y_{ia} = \mu_{k_i}(a) + u_{ia} + \varepsilon_{ia}, \qquad u_{ia} = \rho^g_{k_i}\, u_{i,a-g} + \eta_{ia}, \quad \eta_{ia} \sim N\Big(0,\ \sigma^2 \frac{1-\rho^{2g}_{k_i}}{1-\rho^2_{k_i}}\Big), \quad \varepsilon_{ia} \sim N(0, \sigma^2_\varepsilon). \tag{15}$$

  Shares .25/.47/.29, persistence .87–.96. The worst class now starts near average and falls away.
- On h and frailty the healthiest class collapses at the bound: one reason to use θ.

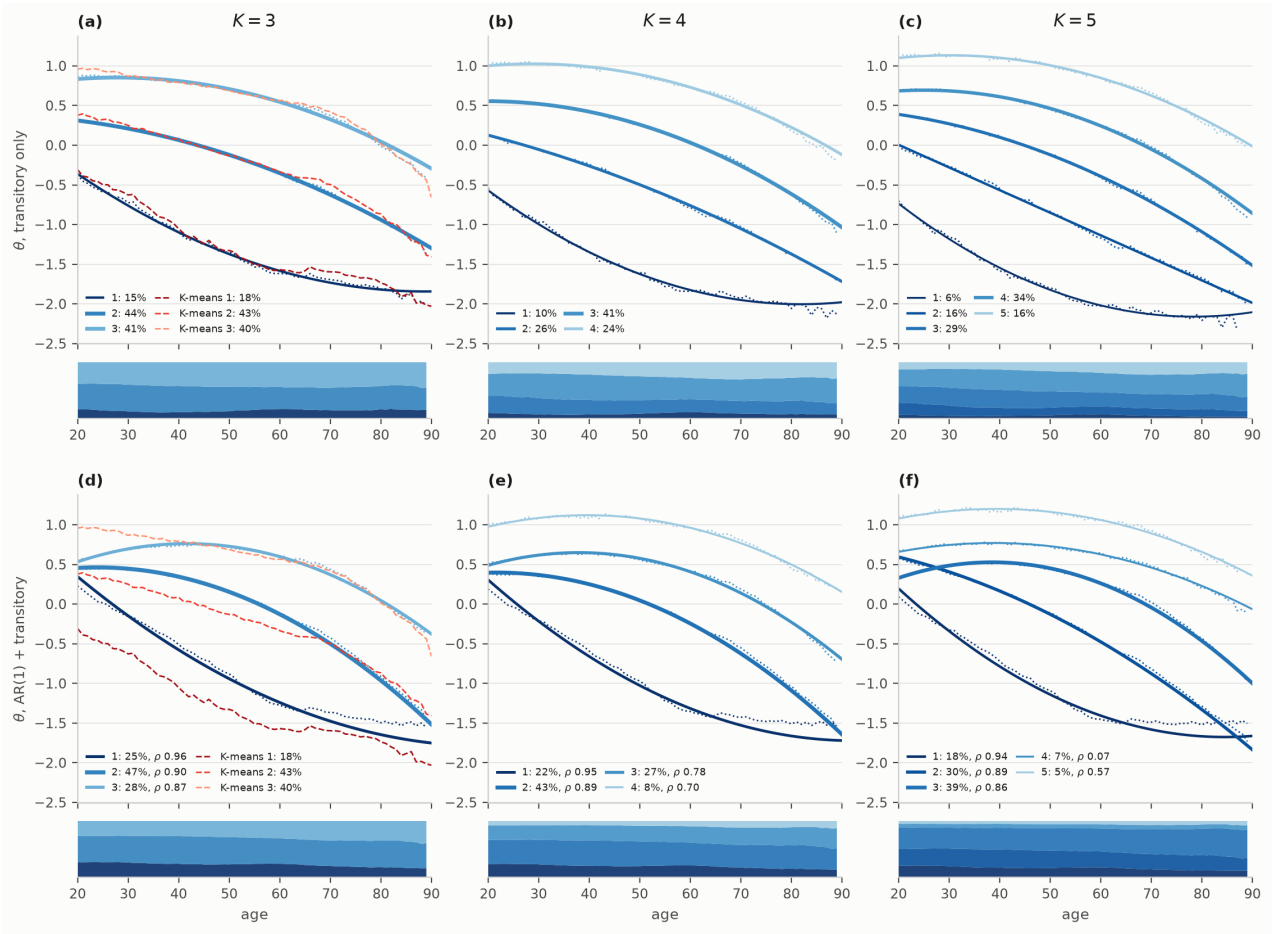


Figure 6: Quadratic growth mixtures on θ: transitory only (a)–(c) and persistent–transitory (d)–(f), at $K = 3, 4, 5$. Line width is proportional to the class share; dotted lines are the observed class means; red dashed lines ($K = 3$) are the partial K-means types; the strips show class composition by age.

- Figure 6: Bayesian model with transitory only shocks is almost identical to K-means.
- $K = 4, 5$: under the transitory-only specification each new class is a sicker group; under the persistent–transitory specification, small healthy groups.

- There is allways a "bad" group that looks similar under every method, and mostly the same people (74–85% overlap).
- The stochastic structure changes how they got there: level-based methods overstate early separation and understate decline, and decline is what prevention prices.
- This group drives the level–slope reversal: drop it and the reversal disappears.

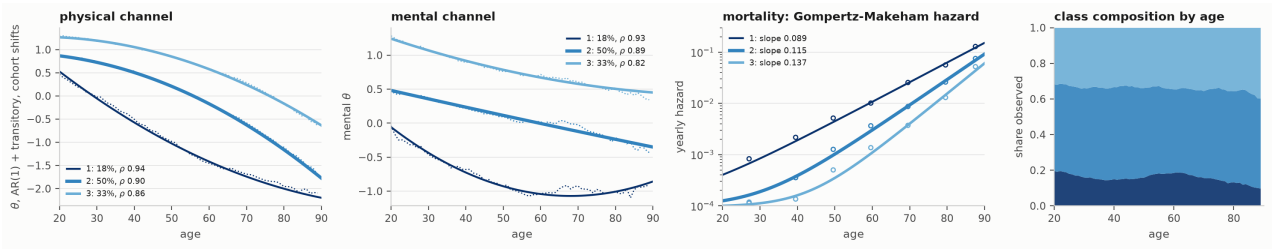


Figure 7: Multidimensional persistent–transitory mixture ($K = 3$) with birth-decade shifts: physical and mental class paths (dotted: observed class means), Gompertz–Makeham class hazards (circles: crude death rates), and class composition by age. Paths are at the 1950s cohort.

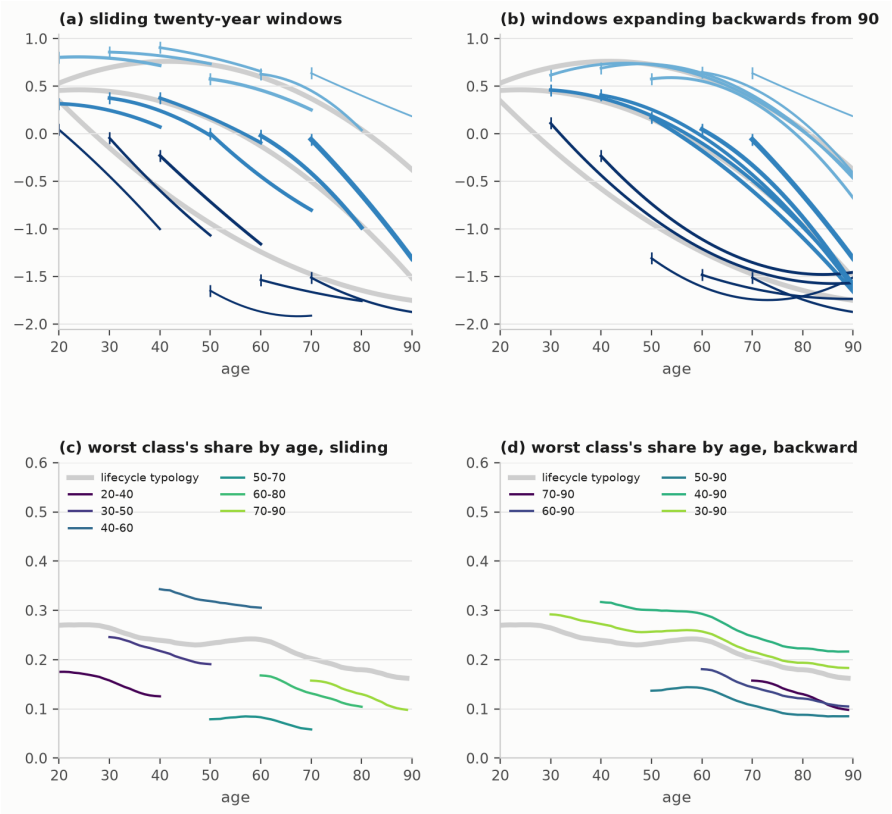


Figure 8: Persistent–transitory classes ($K = 3$, θ) refitted on (a) sliding twenty-year and (b) backward-expanding age windows, against the lifecycle fit in grey (line width proportional to the class share); (c), (d) the worst class's share of the person-waves observed at each age, one line per window.

- Figure 7 shows multidimensional model (physical, mental, Gompertz–Makeham mortality, shared classes): the physical typology survives; mental health is faster and noisier; the worst class's hazard

is 11.5× the best at 55 and 3.2× at 85.

- Cohort shifts leave the typology unchanged but can't be read as clean cohort effects .
- Death is hard to predict: type adds a little over age and less than current health.
- Life expectancy from 25: 47.7/57.2/62.2 years; healthy LE separates the classes far more (17.5 vs 59.2).
- Figure 8 shows the clustering over different windows. A window finds the bad type only if it contains the 40–60 divergence, so this age range is crucial for identifying health types?

4.4 Types against observables

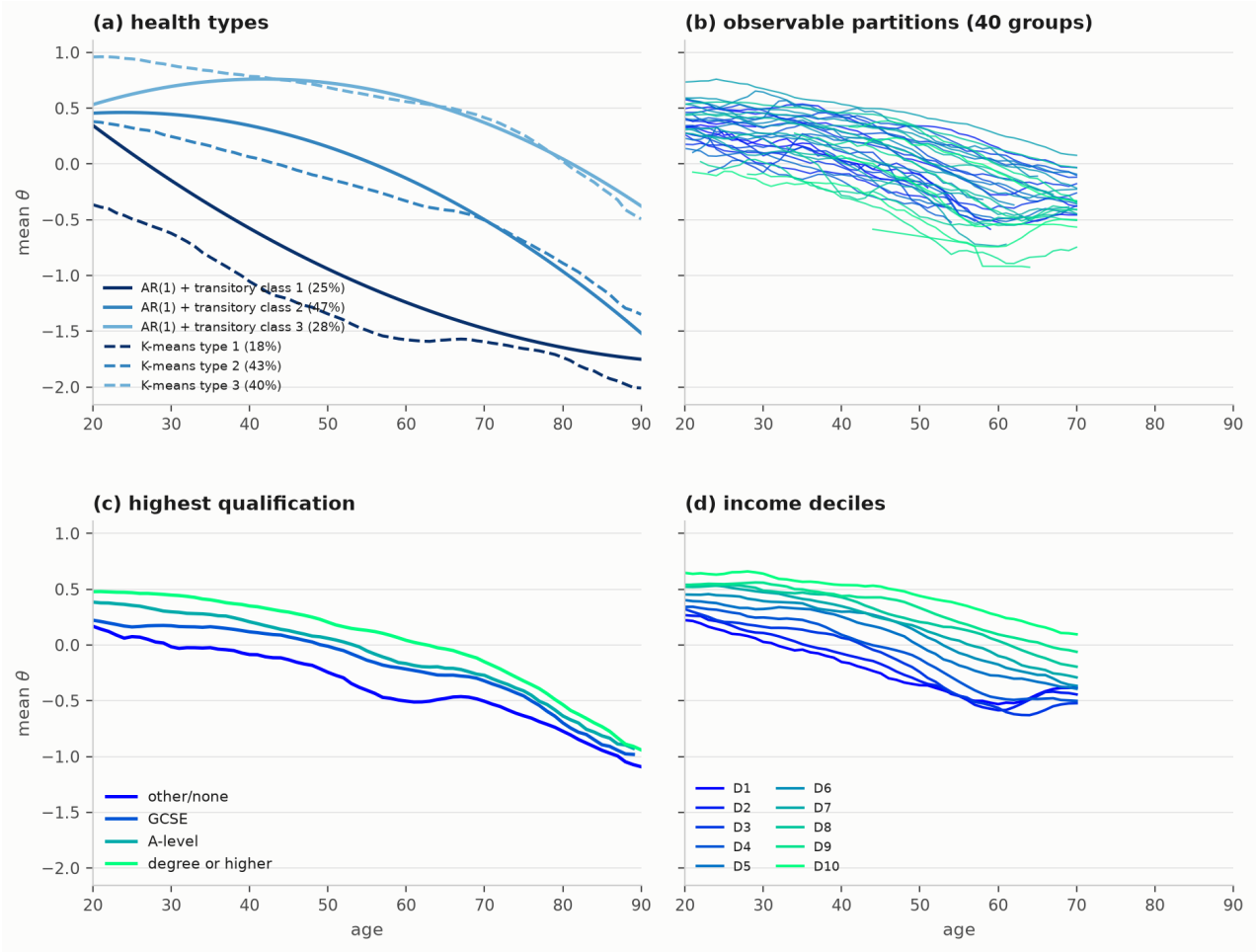


Figure 9: Mean θ by age for (a) the persistent–transitory classes (solid) and partial K-means types (dashed), (b) 40 cells of education, sex, ethnicity and income, (c) education, (d) income decile; (b) and (d) stop at 70.

- Health types explain much more of the health distribution that other observables.
- Gradients by education and income exist and run the expected way (see also Figure 12).
- But even 40 observable cells span half the range the types do; between-group variance shares 3–9% against 48–65% for types; ARI with types 0.01–0.03.

4.5 Predictive value

- Predicting ahead (last two rows held out): observables add nothing; current health and the class add about the same; the model forecast does a little better.
- Predicting type (first two rows held out): observables barely predict it; one health reading is most of the value; health is the weakest guide to type when young.
- Headline: health is what matters, the model helps a bit, observables are almost irrelevant.

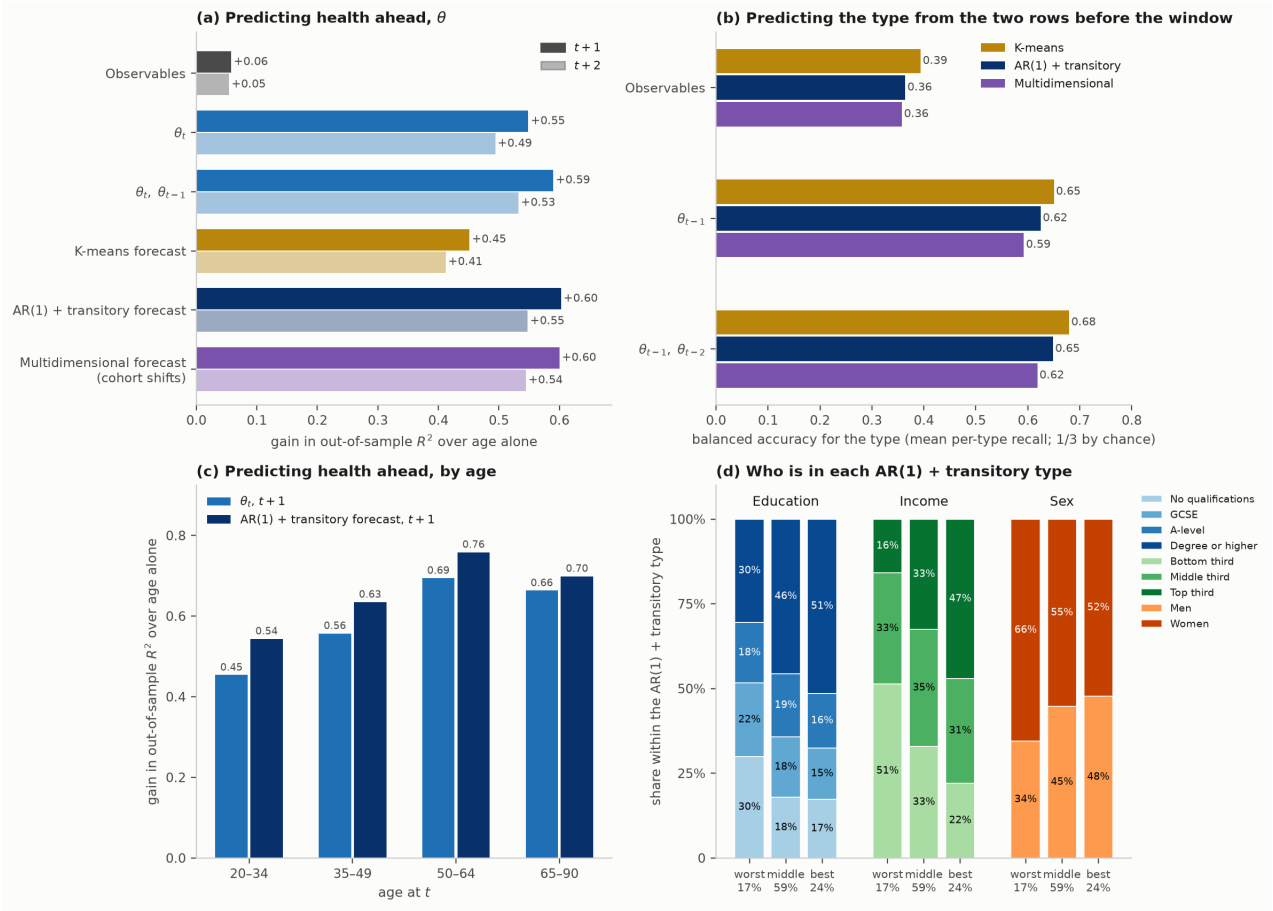


Figure 10: Out-of-sample prediction on θ: (a) gain in R^2 over age alone in predicting θ one and two waves ahead, and (c) the same by age; (b) balanced accuracy in predicting the type from the two rows before the fitted window; (d) the persistent–transitory types by education, income and sex.

5 Prevention exercises

- Priced on the $K = 3$ persistent–transitory univariate physical classes with the estimated cubic cost, undiscounted.
- We can do all of this for the multidimensional, but we would still need to add an extra step on how mortality within a class changes with prevension in order to have Figure 12.
- Expected total cost is £67,500 per person from 25; 63% mean path, 16% type differences, and 21% unpreventable deviations. Note that because costs are non-linear, the linear approximation to returns is only accurate for small μ, so for $\mu = 1$ we use the exact return.

- The exact saving from uniform prevention $P(r)/C(r)$ at $\mu = 1$ is 65% at 25, 52% at 50, 35% at 70 of the total bill. The later the start, the more is already determined.
- The worst class returns 2.3× the uniform amount from 25 because its gap and decline coincide; the best class a tenth.
- For a 10% preventative intervention, returns are highest in the unhealthiest class until age 54, but in older ages it becomes more valuable to target the middle class.
- Figure 11(c) shows tyhat savings are back-loaded (half by 72 for the worst class, 82 middle, 86 best), so discounting would substantially reduce the returns to intervening early.
- Figure 11(d) shows what to target on: age first; then a health reading (£9,100 per treated person at 10% coverage) beats observables (£8,300); the estimated class gets £10,800. Observing current health closes about half the gap from age to class.

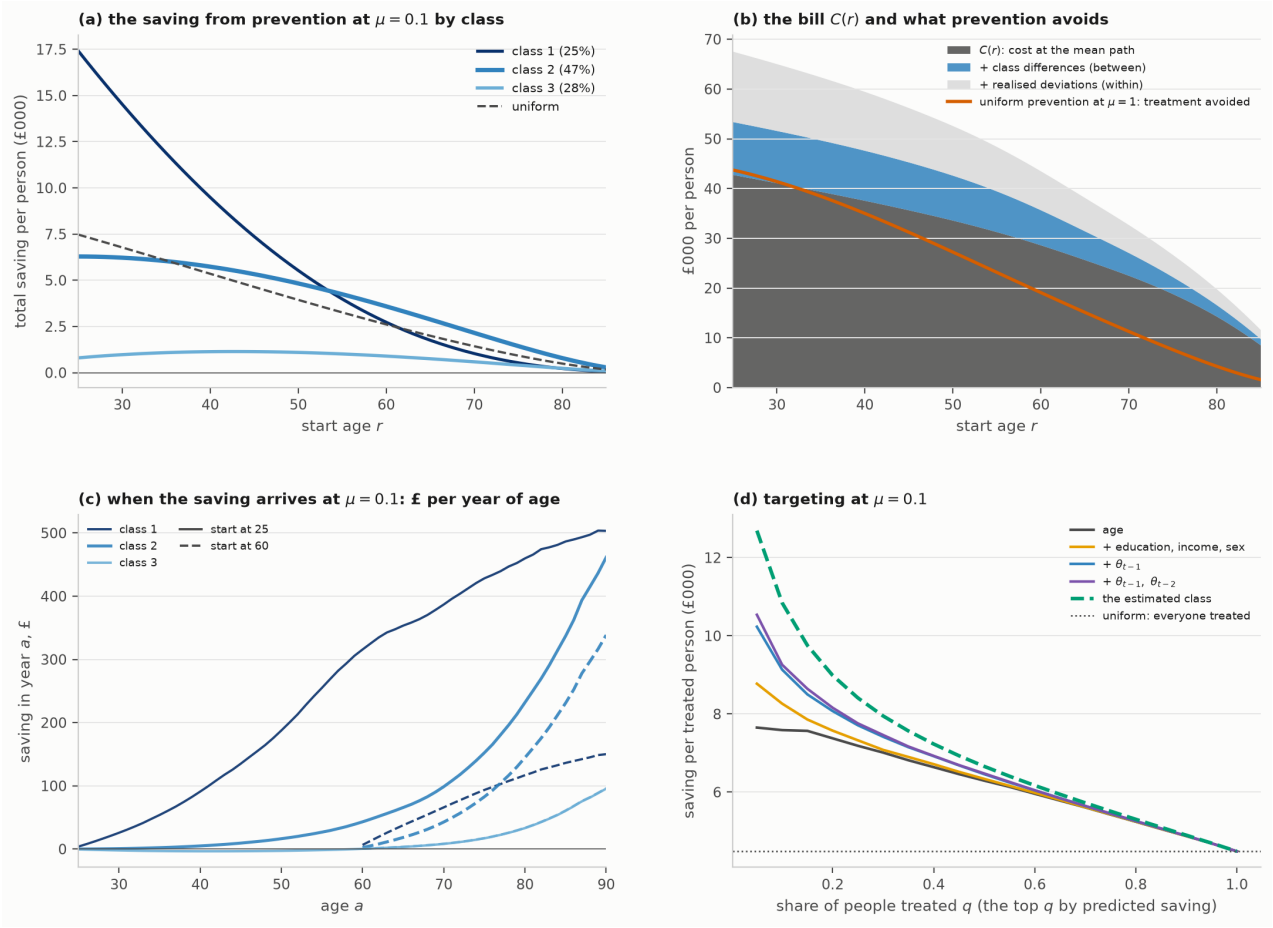


Figure 11: Prevention in pounds on the $K = 3$ persistent–transitory classes on θ, to age 90 and undiscounted; μ is the share of each class's future decline held back. (a) Saving at $\mu = 0.1$ by class and start age (dashed: uniform); (b) the bill $C(r)$, split into the mean path, class differences and unpreventable deviations, with the uniform saving at $\mu = 1$; (c) when the saving arrives, for starts at 25 and 60; (d) saving per treated person when the top q are targeted on what is known before the fitted window.

- Figure 12 looks at implications for inequality.
- Prevention adds more healthy years than years of life.

- Most lifespan inequality is within type (89%) but types explain 30-40% of the education and income gradients.
- Targeting the worst class from 50 halves the LE gap between types (9.5 to 4.6 years); the same budget spread evenly does not. In fact uniform prevention actually slightly exacerbates inequalities as it works against the convergence.
- Targeting the worst class also more than halves the education and income LE gradients the types carry.

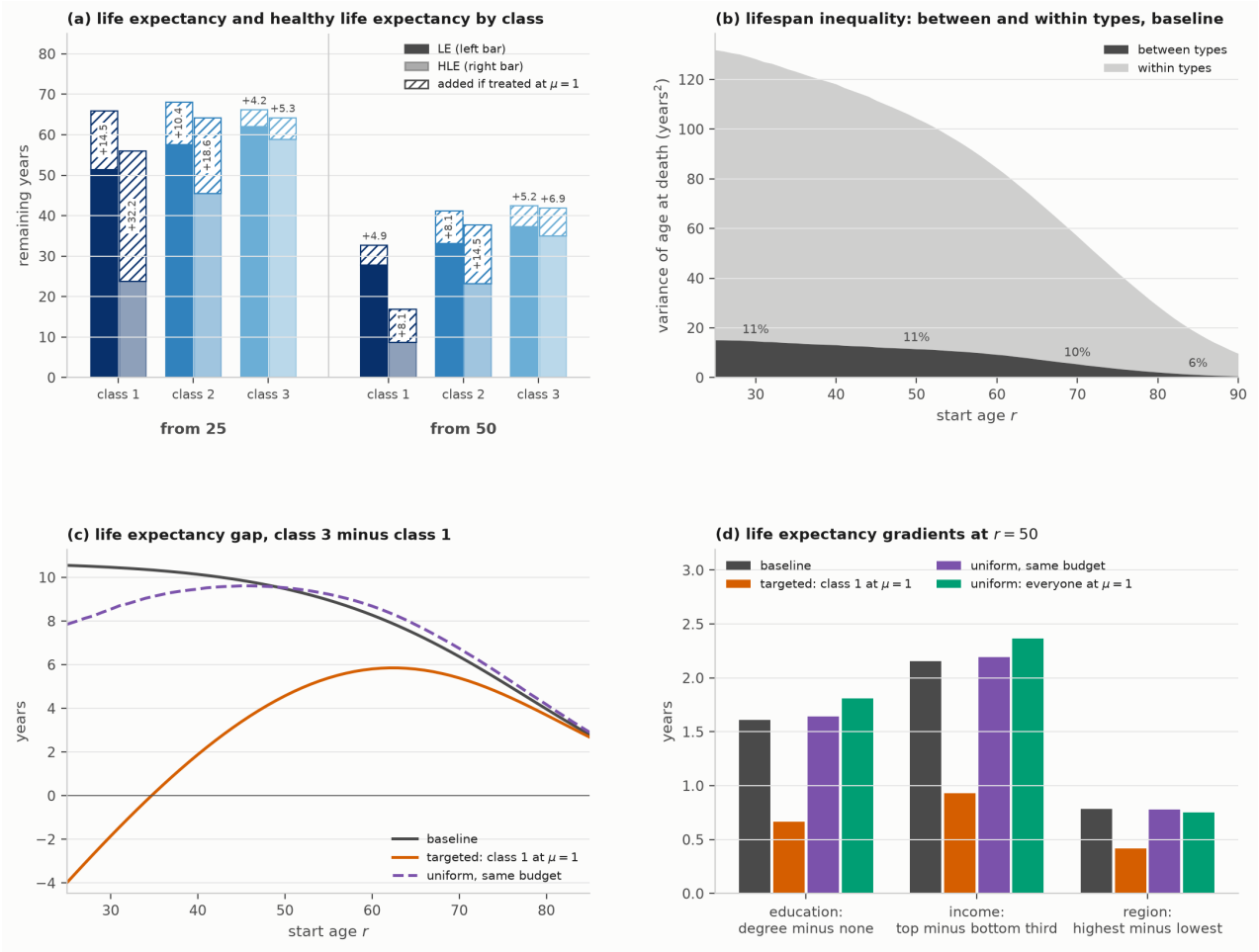


Figure 12: Prevention in life years on the $K = 3$ persistent–transitory classes on θ: (a) LE and HLE by class from 25 and 50, hatched the gain if the class alone is treated at $\mu = 1$; (b) variance of age at death, between and within types; (c) LE gap between the best and worst class and (d) LE gradients at 50 by education, income and region, at baseline, targeting class 1, and under uniform prevention.

6 Conclusions

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